Musterlösung - Absorption in Aluminium

a) $I(x) = I_0 \cdot e^{-\mu_{Al} \cdot x}$ "Absorptionsgesetz"

$\Leftrightarrow \quad I_D = I(D) = I_0 \cdot e^{-\mu_{Al} \cdot D}$

$\Leftrightarrow \quad \frac{I_0}{I_D} = e^{\mu_{Al} \cdot D}$

$$\mu_{AL} = \frac{1}{D} \cdot \ln\left(\frac{I_0}{I_D}\right)$$

$$\mu_{Al} = \frac{1}{5\,mm} \ln\left(\frac{100\ W/m^2}{0{,}34\ W/m^2}\right) \approx \underline{1{,}14\ mm^{-1}}$$

b) Halbwertsdicke d_H

$$I(d_H) \stackrel{!}{=} I_0/2$$

$$\cancel{I_0/2} = \cancel{I_0} \cdot e^{-\mu_{Al} \cdot d_H}$$

$$2 = e^{\mu_{Al} \cdot d_H}$$

$$\ln(2) = \mu_{AL} \cdot d_H$$

$$d_H = \frac{\ln(2)}{\mu_{AL}} \approx \frac{\ln(2)}{1{,}14\ mm^{-1}} \approx \underline{0{,}61\ mm}$$